Conductor-Shell Fourier Energy and Local Refinements for Primitive Quartic Torus Packets

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Abstract

We analyze the local conductor Fourier mechanism in the source claim of equidistribution for arbitrary-order packets in primitive totally real quartic fields. Self-dual trace measure, quadratic phase and finite trace-fiber normalization are kept explicit. We prove an exponent-sensitive shell period, removing the exceptional shift loss at 5, and retain the full negative Smith mass and Walsh energy in local bounds. Exact scalar inversion gives minimum density 7/8 and explicit finite Euler factors. Orthogonal shell energy yields a quantitative local L^2 truncation estimate. Conditional on the source's cubic-resolvent and order-scale inputs, the profile cap reduces from X^{371} to X^{49} by retaining the discriminant-index relation. Previously developed moment and entropy bookkeeping is specialized to this quartic setting without transferring prime-degree classification. Exterior multiplicity, global nonscalar savings, the weighted sieve and homogeneous obstruction exclusion remain separate obligations. Contribution assessment: C3/5, substantive proved local refinements and conditional profile compression, without global arithmetic or equidistribution certification.

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1 Scope and quartic-specific dependencies

This report concerns catalogue ID 35, applicability L3, and the October 5, 2026 source *Equidistribution of primitive quartic torus packets for arbitrary orders* [1]. The source claims equidistribution without escape of mass for complete volume-weighted packets of invertible ideals of any order in a primitive totally real quartic field. We distinguish that global claim from the local results proved here. In degree four, positive entropy does not by itself identify Haar measure: a proper homogeneous obstruction with two coordinate blocks of dimension two must still be excluded. The prime-degree conclusion of M034 is not transferred.

Let K be totally real of degree four with no proper intermediate field, $R = \mathcal{O}_K$, and $\mathcal{O} \subset R$ a full-rank unital order. Write

$$D = |\operatorname{Disc} K|, \quad q = [R : \mathcal{O}], \quad X = q\sqrt{D} = \sqrt{|\operatorname{Disc} \mathcal{O}|}, \quad \kappa = \operatorname*{Res}_{s=1} \zeta_K(s).$$

The ambient space is $X_4 = \mathrm{SL}_4(\mathbb{Z})\backslash \mathrm{SL}_4(\mathbb{R})$ of covolume-one row lattices, with right positive-diagonal action A_4. Its Haar measure uses $du_1 du_2 du_3$ and $u_4 = -u_1 - u_2 - u_3$. The source packet uses all ordinary invertible ideal classes and all sixteen coordinate sign matrices, each distinct orbit once, weighted by orbit Haar volume. A determinant-negative sign matrix acts on the underlying lattice; a positive basis then represents it in X_4. This is not an arbitrary full local module packet or a single-orbit claim.

Source component	Function in the argument
Introduction, packets	Statement, local principality, complete class average, real signs and exact normalization.
Ordinary, rigidity	Weighted ideal count, tightness and compact ball bounds; dominated-component entropy and the two-block obstruction.
Exterior	Cubic resolvent field, opposite-coordinate products, exterior multiplicities and stabilizer-order discriminant bounds.
Local Fourier	Self-dual measures, quadratic transform, conductor shell decay, inverse support and scalar trace marginal.
Additive	Finite shell profiles, Poisson summation, arithmetic nonscalar bounds, scalar term and relative-error conversion.
Sieve, conclusion	Unramified multiplicative weights, planar exterior count, exterior unfolding and exclusion of the obstruction.

The source's exterior construction, multiplicity majorant, global order scale, nonscalar arithmetic estimate, sieve and homogeneous classification remain imported unless explicitly listed as hypotheses of a proved local consequence. Reading their statements or selected steps does not certify their complete proof chains. The read-scope ledger distinguishes full literal reads, selected intervals, structural searches and unread ranges.

2 The exact local quadratic setting

Fix a rational prime p. Let $E = \prod_{\nu=1}^{r} E_\nu$ be a cubic étale $\mathbb{Q}_p$-algebra, $R_E = \prod R_\nu$, with uniformizers π_ν, integer valuations v_ν, ramification indices e_ν, and residue cardinalities $q_\nu = p^{f_\nu}$. Thus $\sum e_\nu f_\nu = 3$. Write $\mathfrak{d}_E = \prod \mathfrak{q}_\nu^{d_\nu}$ for the different. Let W be free of rank two over E, with action ρ, and let B be a nondegenerate symmetric $\mathbb{Q}_p$-bilinear form for which ρ is self-adjoint. Let $L \subset W$ be a full $\mathbb{Z}_p$-lattice, self-dual for B, and assume $Q_1(x) = B(x,x)/2$ is integral on L. Define

$$Q_A(x) = \tfrac{1}{2} B(\rho(A)x, x), \qquad \operatorname{Tr}(A\Psi(x)) = Q_A(x), \qquad \mathcal{C} = \{A \in E : \rho(A)L \subset L\}, \quad \mathcal{C}^\sharp = \mathbb{Z}_p + 2\mathcal{C}.$$

These are abstract local hypotheses. Identifying them with the quartic exterior lattice and its cubic resolvent uses the source's specialized construction, not a generic coordinate change.

Choose ψ_p of conductor $\mathbb{Z}_p$. On W use the self-dual measure for B, so $\operatorname{vol}(L) = 1$. On E use self-dual trace measure, so

$$\operatorname{vol}(R_E) = \prod_\nu q_\nu^{-d_\nu/2}, \qquad \mathfrak{a}^\perp = \mathfrak{a}^{-1}\mathfrak{d}_E^{-1}, \qquad \operatorname{vol}(\mathfrak{a})\operatorname{vol}(\mathfrak{a}^\perp) = 1. \tag{1}$$

In particular R_E does not generally have volume one. The law of $u\Psi(x)$, with independent additive-uniform $x \in L$ and Haar-uniform $u \in \mathbb{Z}_p^\times$, has characteristic function

$$\gamma_p(A) = \mathbb{E}_u \int_L \psi_p(uQ_A(x))\,\mathrm{d}x. \tag{2}$$

We use the source's plus-sign Fourier transform and minus-sign inverse. The source proves that this law has a locally constant finite density f_p^* away from vanishing component coordinates, supported on $(\mathcal{C}^\sharp)^\vee$. For support alone, the calculation is immediate: $\operatorname{Tr}((a+2c)\Psi(x)) = aQ_1(x) + B(\rho(c)x, x) \in \mathbb{Z}_p$ for $a \in \mathbb{Z}_p, c \in \mathcal{C}$. It also proves $\gamma_p(A + c) = \gamma_p(A)$ for $c \in \mathcal{C}^\sharp$.

For an integral ideal $\mathfrak{h} = \prod \mathfrak{q}_\nu^{h_\nu}$ put

$$\mathcal{S}_\mathfrak{h} = \prod_{h_\nu = 0} R_\nu \times \prod_{h_\nu > 0} \pi_\nu^{-h_\nu} R_\nu^\times, \quad H = \mathrm{N}\,\mathfrak{h}, \quad p^g = [\mathfrak{h} : \mathfrak{h} \cap \mathcal{C}].$$

Let $\gamma_\mathfrak{h} = \mathbf{1}_{\mathcal{S}_\mathfrak{h}}\gamma_p$ and $K_\mathfrak{h} = \check{\gamma}_\mathfrak{h}$. Every shell is invariant under R_E-translation; its zero extension retains the $\mathcal{C}^\sharp$-period. Both functions are Schwartz–Bruhat, and $\operatorname{supp} K_\mathfrak{h} \subset (\mathcal{C}^\sharp)^\vee$. A shell inverse need not be nonnegative, even though the complete density is nonnegative.

3 Exact scalar trace law

Here assume the additional integral split identity $Q_1 = x_1y_1 - x_2y_2 + x_3y_3$ in coordinates on L. For $a \in \mathbb{Q}_p$ scalar, integration in each y_i gives

$$\int_{\mathbb{Z}_p^2} \psi_p(uaxy)\,\mathrm{d}x\,\mathrm{d}y = \begin{cases} 1 & a \in \mathbb{Z}_p, \\ p^{-k} & v_p(a) = -k < 0. \end{cases}$$

Thus $\gamma_p(a) = 1$ or p^{-3k} respectively, before the unit average. This derivation is valid at $p = 2$ without dividing by two inside $\mathbb{Z}_p$. Denote the trace density with respect to Haar probability on $\mathbb{Z}_p$ by $d_p(t)$.

Proposition 3.1 (Closed scalar density)**.** *For $t \in \mathbb{Z}_p$ with $v_p(t) = m < \infty$,*

$$d_p(t) = (1 - p^{-3}) \sum_{j=0}^{m} p^{-2j}, \qquad d_p(0) = \frac{1 - p^{-3}}{1 - p^{-2}} = 1 + \frac{1}{p(p+1)}. \tag{3}$$

Consequently $1 - p^{-3} \le d_p(t) \le 1 + 1/(p(p+1))$, with equality at units and at zero respectively. The uniform minimum over all primes and inputs is $7/8$, instead of the source's lower estimate $5/6$.

Proof. The nonconstant Fourier coefficients have total absolute sum $\sum_{k\ge1}(p^k - p^{k-1})p^{-3k} = 1/(p(p+1))$, so compact-group inversion converges uniformly. The sum of characters of exact denominator p^k, evaluated at t, equals $p^k - p^{k-1}$ if $k \le m$, equals $-p^m$ if $k = m + 1$, and vanishes otherwise.

This follows by subtracting the sums over all characters of denominator dividing p^{k-1} from those dividing p^k. Therefore

$$d_p(t) = 1 + (1-p^{-1})\sum_{k=1}^{m} p^{-2k} - p^{-2m-3}.$$

Summing the geometric series gives (3). At zero every character is one. Positivity and the two extreme values follow directly. □

Corollary 3.2 (Finite and infinite scalar products)**.** *For every finite set S and every $t_p \in \mathbb{Z}_p$,*

$$\zeta(3)^{-1} \le \prod_{p\in S} d_p(t_p) \le \zeta(2)/\zeta(3).$$

For a fixed nonzero integer m, the absolutely convergent all-prime product is

$$\prod_p d_p(m) = \frac{\sigma_{-2}(|m|)}{\zeta(3)}, \qquad \sigma_{-2}(N) = \sum_{d|N} d^{-2}.$$

For a finite S the exact formula instead is $\prod_{p\in S}(1-p^{-3})\prod_{p\in S}\sum_{j=0}^{v_p(m)} p^{-2j}$. If S contains every prime divisor of m, its second product is $\sigma_{-2}(|m|)$.

Proof. Multiply the local lower and upper bounds and compare with the Euler products, whose factors are respectively below and above one. For fixed $m \neq 0$, only finitely many valuation sums differ from one, and their product is the stated divisor sum. Absolute convergence follows from $\sum_p p^{-2} < \infty$. These universal bounds are infimum and supremum limits, not assertions that a finite S attains them. □

The source's additive main term is $A_\phi l^{-2}\prod_{p\in S} d_p(m)$. It must retain its actual finite S. Neither the closed formula nor a positive scalar lower bound supplies the nonscalar power saving.

4 Retaining the Smith mass and optimizing support

Put $\iota = \mathbf{1}_{\{p=2,3\}}$. For $A \in \mathcal{S}_{\mathfrak{h}}$ some $s \in \{0,1,2,3\}$ makes $\alpha = A + s$ invertible, with

$$\beta = \alpha^{-1} \in p^{-\iota}\mathfrak{h}, \qquad p^{-3\iota}H \le |\operatorname{N}\alpha|_p \le H. \tag{4}$$

Indeed a negative component valuation is unchanged by a scalar integer. For an integral component and $p \ge 5$, at most one of four distinct scalar residues is forbidden, and there are at most three components. For $p = 2, 3$, two distinct candidates have p-adic difference valuation at most one; thus at most one can give component valuation greater than e_ν. Choose a candidate not excluded by any component. Multiplying the resulting component norm bounds proves (4). This also removes $p = 5$ from the source's exceptional shift constant.

The quadratic transform used below is

$$\int_L \psi_p(uQ_\alpha) = \omega(u,\alpha)|\operatorname{N}\alpha|_p^{-1}\int_L \psi_p(-u^{-1}Q_\beta), \qquad |\omega| = 1, \quad \omega(u,\alpha r^2) = \omega(u,\alpha). \tag{5}$$

Here $r \in E^\times$. For clarity, completing the square gives the distributional Fourier transform of $\psi(B(Px,x)/2)$ as a constant times $\psi(-B(P^{-1}y,y)/2)$. Integrating first on $p^{-M}L$ and pairing with a compactly supported test function makes this identity stable for large M. Differencing the squared Gaussian integral then gives its absolute value $|\det P|_p^{-1/2}$. Pairing with $\widehat{\mathbf{1}_L} = \mathbf{1}_L$ proves

the lattice identity. A change of variables shows invariance of its normalized phase under isometry. For $P = u\rho(\alpha)$ the determinant is $u^6\,\mathrm{N}(\alpha)^2$; $\rho(r)$ effects the stated isometry. This is the standard quadratic Fourier identity, with the source's normalization, not a new theorem about Weil indices [5].

Proposition 4.1 (Full inverse-index information)**.** *Let $p^{a_1},\dots,p^{a_6}$ be the Smith divisors of $\rho(\beta)$ relative to L, and set*

$$R_\beta = L \cap \rho(\beta)^{-1}L, \quad k(\beta) = \sum_i \max(0,-a_i), \quad j = \max(0,-\min_i a_i).$$

For each unit u the squared absolute value of the inverse integral in (5) *is $p^{-k(\beta)}$ if $Q_\beta(R_\beta) \subset \mathbb{Z}_p$, and is zero otherwise. At odd p the integrality condition always holds. In particular*

$$|\gamma_p(A)| \le p^{3\iota} H^{-1} p^{-k(\beta)/2}, \qquad \|K_{\mathfrak{h}}\|_\infty \le p^{3\iota}\,\mathrm{vol}(R_E). \tag{6}$$

Proof. Writing $I = \int_L \psi_p(-u^{-1}Q_\beta(x))\,\mathrm{d}x$, additive orthogonality on the self-dual lattice gives

$$|I|^2 = \int_{R_\beta} \psi_p(-u^{-1}Q_\beta(h))\,\mathrm{d}h.$$

On R_β the polar term $B(\rho(\beta)h,k)$ is integral. Consequently the integrand is a character of its additive compact group. Its integral is either its volume or zero, according as that character is trivial or not. The Smith normal form gives volume $p^{-k(\beta)}$. For odd p, $B(\rho(\beta)h,h)/2$ is integral; at $p=2$ this step must not be assumed. Scalar shifting leaves the original integrand unchanged because $Q_1(L)$ is integral. Now apply (5), (4), and the triangle inequality for the unit average. Finally the shell has volume at most $H\,\mathrm{vol}(R_E)$. □

The source already displays the full index before replacing it by p^j. Thus (6) is a retained-information refinement, not a new general Gaussian bound. Its advantage can be strict when several Smith directions are negative. It does not prove how often such directions occur in the global arithmetic sum; phases can still cancel in the unit average.

Theorem 4.2 (Exponent-sensitive shell period)**.** *Let p^c be the exponent of the finite group $\mathfrak{h}/(\mathfrak{h}\cap\mathcal{C})$, with $c=0$ for the zero group. Then $0 \le c \le g$, and the zero-extended shell function has period $p^N\mathfrak{h}^{-1}$ for*

$$N = \begin{cases} \max(1,c), & p \ge 5, \\ c+2, & p = 3, \\ \max(4,c+3), & p = 2. \end{cases} \tag{7}$$

In particular

$$\operatorname{supp} K_{\mathfrak{h}} \subset p^{-N}\mathfrak{h}\mathfrak{d}_E^{-1}. \tag{8}$$

This is a sufficient support exponent, not an assertion of minimality.

Proof. By definition $p^c\mathfrak{h} \subset \mathcal{C}$. Take $t \in p^N\mathfrak{h}^{-1}$ and retain the scalar s in (4). The following three inequalities suffice:

$$N \ge 1, \qquad N - \iota \ge \begin{cases} 1 & p \ne 2, \\ 3 & p = 2, \end{cases} \qquad N - 2\iota \ge c + v_p(2). \tag{9}$$

They hold for (7). Translation by t preserves the shell. For $\zeta = t\beta$ one has

$$\zeta \in p^{N-\iota} R_E, \quad \alpha + t = \alpha(1+\zeta), \quad (\alpha+t)^{-1} - \beta = -\frac{t\beta^2}{1+\zeta} \in p^{N-2\iota}\mathfrak{h} \subset 2\mathcal{C}.$$

At odd primes every principal unit here is a square. At $p = 2$, $v_\nu(\zeta) \geq 3e_\nu > 2v_\nu(2)$ gives a square by Hensel's lemma, including ramified factors. Norms and Gaussian phases in (5) are unchanged. The inverse quadratic integral is unchanged since $Q_{2a}(L) = B(\rho(a)L, L) \subset \mathbb{Z}_p$ for $a \in \mathcal{C}$. Thus the original integral, separately for every u, has the stated period. Its zero extension does too, because the shell is translation-invariant. Taking the trace annihilator and using (1) proves (8). □

For $p > 5$ and $g > 0$ this replaces the source's $4(g+1)$ by $c \leq g$. Unlike the group order, the group exponent detects a conductor defect spread over several directions. No exterior-lattice multiplicity or cubic-resolvent identification has been weakened.

5 Boundary kernels and square-integrable density

Proposition 5.1 (Odd-prime boundary estimate)**.** *Suppose p is odd and $g = 0$. Support requires $z_\nu \in \mathfrak{d}_\nu^{-1}$ for $h_\nu = 0$, and $z_\nu \in \mathfrak{q}_\nu^{h_\nu - 1}\mathfrak{d}_\nu^{-1}$ otherwise. Let B_z denote the positive-h_ν components where $z_\nu \notin \mathfrak{q}_\nu^{h_\nu}\mathfrak{d}_\nu^{-1}$. On this support,*

$$|K_{\mathfrak{h}}(z)| \leq \operatorname{vol}(R_E) \prod_{\substack{h_\nu > 0 \\ \nu \notin B_z}} (1 - q_\nu^{-1}) \prod_{\nu \in B_z} (q_\nu^{-2} + q_\nu^{-1})^{1/2}. \tag{10}$$

Proof. Here $\mathfrak{h} \subset \mathcal{C}$. Component idempotents with $h_\nu = 0$ split L orthogonally into self-dual summands; their integral quadratic factors contribute one, using that 2 is a unit. On the remaining combined summand the inverse frequency lies in $\mathfrak{h} \subset \mathcal{C}$, so the inverse integral is one. The shell function is therefore H^{-1} times a function of modulus at most one on the product of the two unit square classes in each positive component. This assertion follows from isometry invariance of the Gaussian phase; the isometry need not preserve L.

Expand that bounded function in normalized Walsh characters with coefficients b_ε. Finite Parseval gives $\sum_\varepsilon |b_\varepsilon|^2 \leq 1$, including coupling from the common unit u. In a positive component the normalized inverse of the trivial character is

$$\operatorname{vol}(R_\nu)(\mathbf{1}_{\mathfrak{q}_\nu^{h_\nu}\mathfrak{d}_\nu^{-1}} - q_\nu^{-1}\mathbf{1}_{\mathfrak{q}_\nu^{h_\nu - 1}\mathfrak{d}_\nu^{-1}}).$$

The quadratic-character inverse vanishes in the interior and outside the displayed support; on the boundary its magnitude is $\operatorname{vol}(R_\nu) q_\nu^{-1/2}$. Indeed the finite-field Gauss sum has squared magnitude q_ν: expand its square, put $a = tb$, and use the additive-character sum over nonzero b, equal to $q_\nu - 1$ at $t = 1$ and -1 otherwise. Cauchy–Schwarz in the Walsh coefficients now gives the product of the two-component Euclidean norms. An integral component contributes $\operatorname{vol}(R_\nu)$. This proves (10) and the support assertion. □

This retains the source's boundary decay while replacing its factor 8 by at most $\prod_{\nu \in B_z} \sqrt{1 + q_\nu^{-1}}$, and shows that this part of the argument needs only odd residue characteristic, not $p > 5$. We make no analogous square-class assertion at $p = 2$.

Theorem 5.2 (Local orthogonal energy and truncation)**.** *The law in* (2) *has an* $L^2(E)$ *density* f*. For every finite set* $\mathcal{H}$ *of shells,*

$$\left\| \sum_{\mathfrak{h} \in \mathcal{H}} K_{\mathfrak{h}} \right\|_2^2 = \sum_{\mathfrak{h} \in \mathcal{H}} \int_{\mathcal{S}_{\mathfrak{h}}} |\gamma_p(A)|^2 \, \mathrm{d}A. \tag{11}$$

For every $B_0 \geq 1$ *and fixed* $0 < s < 1$*,*

$$\left\| f - \sum_{H \leq B_0} K_{\mathfrak{h}} \right\|_2^2 \leq p^{6\iota} \operatorname{vol}(R_E) B_0^{-s} \prod_{\nu} (1 - q_\nu^{s-1})^{-1}. \tag{12}$$

Proof. The frequency shells partition E and are disjoint. Plancherel for the self-dual measure proves (11). From (6), the integral for one shell is at most $p^{6\iota} \operatorname{vol}(R_E) H^{-1}$; retaining $p^{-k(\beta)}$ can only decrease it. The sum of H^{-1} over all shells is $\prod_\nu (1 - q_\nu^{-1})^{-1} < \infty$. Hence $\gamma_p \in L^2$. Its inverse is the density of the original probability measure: Fourier inversion against Schwartz–Bruhat test functions identifies the two distributions, so the L^2 representative is nonnegative, integrable, and has mass one. The source's f_p^* agrees with it almost everywhere. Summing the omitted shell energies and using $H^{-1} \leq B_0^{-s} H^{s-1}$ when $H > B_0$ gives (12). □

The source separately obtains finite pointwise shell inversion at inputs whose component coordinates are all nonzero. This theorem instead gives mean-square convergence including neighborhoods of the singular coordinate locus. Neither statement bounds a global sum of density values at arithmetic points merely from its local L^2 norm.

6 A finite trace-fiber energy formulation

This section implements the finite-energy and exact-normalization principles in the private originals [2, 3]. Those proofs give Plancherel and positive Gram identities, not arithmetic cancellation. Here the measure is not Euclidean Fourier measure: it is the self-dual trace measure of (1), followed by uniform probability on a finite quotient. No Mellin-line norm is silently substituted for it.

Let $\Lambda = (\mathcal{C}^\sharp)^\vee$, a compact additive lattice with integral trace. Choose an open subgroup $J \subset \Lambda$ with $\operatorname{Tr} J \subset p^a \mathbb{Z}_p$, $a \geq 1$, and assume $\operatorname{Tr} : \Lambda/J \longrightarrow \mathbb{Z}/p^a\mathbb{Z}$ is onto. This holds in the split scalar setting: $\Psi(L) \subset \Lambda$ has all integral trace values. Put $G = \Lambda/J$, $T = \mathbb{Z}/p^a\mathbb{Z}$. Push the law of $u\Psi(x)$ to a probability μ on G and write $f(x) = |G|\mu(x)$, $d(t) = |T|\mu(\operatorname{Tr} = t)$. All finite norms below use normalized counting measure, and $\widehat{f}(\chi) = |G|^{-1} \sum_x f(x)\overline{\chi(x)}$. The minus convention here is immaterial for squared magnitudes; it is stated to avoid confusing finite coefficients with raw sums.

Proposition 6.1 (Trace projection and residual energy)**.** *Let* P *average over each fiber of* $\operatorname{Tr} : G \to T$*. Then* $Pf = d \circ \operatorname{Tr}$*, and*

$$\mathcal{E} = \|f - Pf\|_2^2 = \sum_{\chi \notin \operatorname{Tr}^* \widehat{T}} |\widehat{f}(\chi)|^2 = |G| \sum_x \mu(x)^2 - |T| \sum_t \mu(\operatorname{Tr} = t)^2. \tag{13}$$

For any complex test function b *on* G*,*

$$|\langle f - Pf, b \rangle| \leq \sqrt{\mathcal{E}} \, \|b - Pb\|_2.$$

If $d(t) > 0$ *and* E_t *is the squared distance from one of the conditional density* $f/d(t)$ *on the fiber, then*

$$\mathcal{E} = |T|^{-1} \sum_t d(t)^2 E_t, \qquad E_t \leq |T| \mathcal{E} / d(t)^2. \tag{14}$$

Proof. All fibers have $|\ker \mathrm{Tr}| = |G|/|T|$ elements, proving the formula for Pf. Averaging a character over the kernel retains it exactly when it factors through T. Thus P is an orthogonal projection and finite Parseval proves the first two equalities; expanding the normalized squared norms proves the last. Orthogonality permits replacing b by $b - Pb$, after which Cauchy–Schwarz applies. Finally split the norm into its equally weighted fibers, factor $d(t)^2$ from each, and use nonnegativity. □

In the split scalar case $d(t)$ is the average of d_p over its residue coset, so it lies between $1 - p^{-3}$ and $1 + 1/(p(p+1))$. This controls normalization but does not make $\mathcal{E}$ small. In particular the factor $|T|$ in (14) prevents an unjustified uniform claim on every trace fiber from a single total-energy estimate. The source's arithmetic nonscalar sum is not this finite residual without further Poisson summation, smoothing and lattice counting.

7 Conditional global shell bookkeeping

We record a concrete downstream consequence without re-proving the quartic resolvent construction. The following are explicit imported inputs from `exterior.tex` and `additive.tex`:

1. A totally real cubic field F, a stabilizer order C_F, and the local quadratic hypotheses of Section 2 for their completions; C_F is maximal outside S.
2. With $I = [R_F : C_F]$ and $\Delta_F = D_F I^2$, $\Delta_F \ll X^{28}$, hence $D_F \ll X^{28}$ and $I \ll X^{14}$. The source's lower bound $\Delta_F \gg X^{1/2}$ is a separate input to its nonscalar argument, not used for our cap.
3. S contains $2, 3, 5$, every prime dividing Dqm, and a fixed set of small primes; $m \neq 0$ is fixed and $\prod_{p \in S} p \ll_m X^2$.
4. The global inputs $z \in F^\times$ are integral outside S and have their three real coordinates in a fixed bounded region.

For a profile $\mathfrak{h}$ supported above S write $H = \mathrm{N}\,\mathfrak{h}$, $G_0 = \prod_{p \in S} p^{g_p}$. The natural maps of finite additive groups give

$$G_0 \leq I \leq H G_0. \tag{15}$$

Locally the first inequality follows by injecting $\mathfrak{h}_p/(\mathfrak{h}_p \cap C_{F,p})$ into $R_{F,p}/C_{F,p}$; the second follows from $[R_{F,p} : C_{F,p}] \leq [R_{F,p} : \mathfrak{h}_p \cap C_{F,p}]$.

Call a rational prime bad for this profile when $p = 2$ or $g_p > 0$. At good primes use Proposition 5.1; let T_z be the product of the residue cardinalities of its boundary components. Each prime has total residue degree at most three, so $T_z \leq \prod_{p \in S} p^3 \ll_m X^6$.

Corollary 7.1 (Reduced profile cap, conditional on the listed inputs)**.** *If $\prod_{p \in S} K_{p,\mathfrak{h}_p}(z) \neq 0$, then*

$$H \ll D_F G_0^3 T_z \ll_m X^{48}. \tag{16}$$

Thus $H \leq X^{49}$ is sufficient for all sufficiently large X. The source's corresponding coarse bounds are $H \ll D_F G_0^{24} T_z$ and the cap X^{371}. Every supported point also lies in a containing fractional ideal $\mathfrak{a}$ with $(\mathrm{N}\,\mathfrak{a})^{-1} \ll D_F G_0^3$.

Proof. At a bad prime the norm cost of enlarging support by p^{-N_p} is p^{3N_p}. Theorem 4.2 gives

$$\prod_{p \text{ bad}} p^{3N_p} \leq 2^{12} 3^6 G_0^3 :$$

for $p \geq 5$ a bad prime has $c_p \geq 1$ and $N_p = c_p \leq g_p$; at 3, $N_p \leq g_p + 2$; at 2, $N_p \leq g_p + 4$. The good support costs exactly one residue-cardinality factor per boundary component relative to $\mathfrak{h}\mathfrak{d}_F^{-1}$. Consequently the containing ideal with these boundary factors has norm $H/(D_F \prod_{\mathrm{bad}} p^{3N_p} T_z)$. For a nonzero element of a fractional ideal, the absolute field norm is at least that ideal's norm. The real bounded region gives $|\mathrm{N}\, z| \ll 1$, proving the first bound in (16). Retain the discriminant-index relation: $D_F G_0^3 \leq D_F I^3 = \Delta_F^{3/2}/\sqrt{D_F} \ll X^{42}$, since $D_F \geq 1$. Hence the second exponent is $42 + 6 = 48$, not the weaker $28 + 3 \cdot 14 + 6$ from treating the factors independently. To obtain $\mathfrak{a}$, discard the positive powers of the prime ideals in the support: at good primes $h_\nu - 1 \geq 0$ for positive h_ν, and at bad primes retain just p^{-N_p} and the inverse different. Its reciprocal norm is at most $D_F \prod_{\mathrm{bad}} p^{3N_p}$, as claimed. □

This cap reduces a finite expansion range, not the modulus of equidistribution. For any fixed $\epsilon > 0$ the number of profiles with $H \leq X^{49}$ is still $O_\epsilon(X^\epsilon)$: Rankin's inequality bounds it by $X^{49u} \prod_{\mathfrak{q}|S} (1 - \mathrm{N}\,\mathfrak{q}^{-u})^{-1}$; for fixed small $u > 0$ the product is $O_{u,v}(X^v)$ for every $v > 0$, by absorbing all sufficiently large prime factors into $p^{v/2}$ and treating the finitely many small primes separately.

The bound $D_F G_0^3$ also improves one isolated small-discriminant, small-index branch of the source's bookkeeping. The containing ideal has minimum nonzero real Euclidean length $\gg (\mathrm{N}\,\mathfrak{a})^{1/3}$ by the norm and arithmetic–geometric mean inequality; disjoint balls therefore bound its points in a fixed box by $O(1 + D_F G_0^3)$. Proposition 5.1 and the bad-prime uniform bound give

$$\left| \prod_{p \in S} K_{p,\mathfrak{h}_p}(z) \right| \ll_\epsilon X^\epsilon T_z^{-1/2} \ll_\epsilon X^\epsilon (D_F G_0^3/H)^{1/2}.$$

Here the finite product of bounded per-prime constants is $X^{o(1)}$, as justified by the same small/large prime split. Conditional on the source's ranges $D_F < X^d$, $G_0 \leq X^e$, and $H \geq X^{\lambda/3-e}$, the original signed profile sum against a bounded test function is thus

$$O_\epsilon\big(X^{-\lambda/6+(3/2)d+5e+\epsilon}\big). \tag{17}$$

The previous exponent from G_0^{24} is $-\lambda/6 + (3/2)d + (73/2)e$. This is only that signed-sum branch; scalar subtraction, large-index inverse congruences, frequency discriminants, and every other branch still require their own bounds. We do not reselect the source's global τ, θ on this basis.

7.1 What remains in the arithmetic interface

For a trace-m smooth box ϕ of two scales r_1, r_2, let $A_\phi = r_1 r_2 \int \Phi$. These are deleted trace coordinates with Jacobian one, not Euclidean surface measure multiplied by $\sqrt{3}$. The imported additive assertion, uniformly over the stated field, order, class and box data, is

$$\sum_{\substack{z \in F,\ \mathrm{Tr}\, z = m \neq 0 \\ z \text{ integral outside } S,\ z \equiv b\ (l)}} \phi(z) \prod_{p \in S} f_p^*(z) = A_\phi l^{-2} \prod_{p \in S} d_p(m)(1 + O(X^{-\theta})), \tag{18}$$

for $r_i \geq X^{-\tau}$, $l \leq X^\theta$, $(l, \prod S) = 1$ and an admissible trace class b. The power l^{-2} comes from the two trace-plane coordinates, although a three-dimensional Poisson frequency factor initially contributes l^{-3}. At the real place the covolume $\sqrt{D_F}$ cancels the finite self-dual volume product $D_F^{-1/2}$, giving global covolume one. Scalar frequencies yield the one-dimensional trace inversion of Section 3. The nonscalar frequencies require genuine uniform arithmetic estimates. An absolute error $O(X^{-\varepsilon_F})$ becomes a relative error of size $O(X^{-\varepsilon_F+2\tau+2\theta})$ after division by $A_\phi l^{-2} \prod_S d_p(m)$; to get the displayed saving requires $\varepsilon_F \geq 2\tau + 3\theta$ (strict inequality leaves slack).

The exterior multiplicity is a weighted count $M(z)$ over ideal representatives and positive-unit orbits of exterior vectors, not a count of distinct z. Its imported local majorant has a factor 32, discriminant and ideal Euler factors, the densities f_p^* at S, and weights b_p outside S. The latter have exact mean $(1-p^{-1})(1-p^{-3})Z_p$ in the source; some means are below one. They cannot be replaced by one or discarded under positivity. The source's weighted sieve and residue comparison must still supply the planar estimate $\sum_{z\in\mathcal{R}} M(z) \ll \kappa\sqrt{D}\,\mathrm{area}(\mathcal{R})$. Neither (13) nor the reduced cap proves it.

8 Ordinary weights and dominated entropy

These two refinements specialize mechanisms already developed in M034; they are not credited as independent new M035 discoveries. Their quartic hypotheses have nevertheless been checked against the present source. The ordinary weights arise from the quartic algebra $K\otimes\mathbb{Q}_p$, not the cubic resolvent algebra of the Fourier sections.

Proposition 8.1 (Correlated quartic local moment)**.** *For the source's containment probability $P_p(\boldsymbol{l}) = \Pr_{t\in R_p^\times}(x\in t\mathcal{O}_p)$ on a valuation shell, put $N_p = p^{\sum f_\nu l_\nu}$ and $Z_p = \prod_{\nu|p}(1-p^{-f_\nu})^{-1}$. For each fixed $0<s<1/4$ and every $\epsilon>0$,*

$$\mathbb{E}_{\mathcal{O}_p} N_p(x)^s \le \frac{1-p^{-1}}{1-p^{4s-1}}, \tag{19}$$

$$\sum_{\boldsymbol{l}} P(\boldsymbol{l}) d(\boldsymbol{l})^{-1+s} \ll_{s,\epsilon} q^{-1+\epsilon}, \tag{20}$$

$$\sum_{d(\boldsymbol{l})>X^{1/8}} P(\boldsymbol{l})/d(\boldsymbol{l}) \ll_\delta q^{-1}X^{-\delta} \quad (0<\delta<1/32). \tag{21}$$

Expectations use additive Haar probability on the full local order. Constants are independent of field, order and its local indices.

Proof. The projection of the order to each R_ν contains scalar $\mathbb{Z}_p$. Its image modulo $\mathfrak{p}_\nu^k$ therefore has at least $p^{\lceil k/e_\nu\rceil}$ elements, and $\Pr(v_\nu(x)\ge k)\le p^{-\lceil k/e_\nu\rceil}$. For $J_\nu = \lceil v_\nu(x)/e_\nu\rceil$ this says $\Pr(J_\nu\ge a)\le p^{-a}$, $a\ge 1$. The tail-sum identity yields

$$\mathbb{E}p^{4sJ_\nu} \le 1+\sum_{a\ge1}(p^{4sa}-p^{4s(a-1)})p^{-a} = \frac{1-p^{-1}}{1-p^{4s-1}}.$$

Apply weighted Hölder with weights $e_\nu f_\nu/4$ to $N_p(x)^s \le \prod_\nu (p^{4sJ_\nu})^{e_\nu f_\nu/4}$. No independence of component projections is needed. The additive measure of a shell in R_p is $1/(Z_pN_p)$, and its fraction lying in $\mathcal{O}_p$ is exactly P_p by unit transitivity. Thus its weighted mass is $Z_pq_p^{-1}\mathbb{E}_{\mathcal{O}_p}N_p^s$. Multiply over $p\mid q$. For fixed $s<1/4$ the extra local factors tend to one as $p\to\infty$, so each is at most p^ϵ for all sufficiently large primes. This proves (20). Markov's inequality gives $q^{-1+\epsilon}X^{-s/8}$; use $q\le X$ and choose $\epsilon = s/8-\delta>0$ after choosing $8\delta<s<1/4$. □

The source uses $s=1/8$ and the weaker tail $X^{-1/128}/q$. Divergence of this particular bound at $s=1/4$ does not disprove an endpoint estimate using more order geometry. The residue comparison $h\zeta_K(1+h)\asymp\kappa$ for $h\le 4/\log D$ and the uniform ideal interval estimate remain imported. In particular $\asymp$ here does not mean a ratio tending uniformly to one: the source integrates an $O(\log D)$ logarithmic derivative, giving $\exp(O(h\log D))$. Primitivity excludes the quadratic subfield in the stated nonnormal Stark exceptional-zero criterion; we do not posit a normality gap.

Proposition 8.2 (Conditional dominated entropy)**.** *Suppose an A_4-invariant probability μ on X_4 satisfies $\mu(xB(r)) \le C_\Omega r^\sigma$ on every compact set of centers, for all sufficiently small r, with $\sigma > 3$. Fix a regular flow $a(t) = \operatorname{diag}(e^{\lambda_1 t}, \dots, e^{\lambda_4 t})$, $\sum \lambda_i = 0$, and $\lambda = \min_{i \ne j} |\lambda_i - \lambda_j|$. Assuming the ELMV tube criterion and entropy decomposition formula, every invariant probability $\xi \le c\mu$, $c < \infty$, has*

$$h_\xi(a(1)) \ge \lambda(\sigma - 3)/2.$$

The same lower bound holds in almost every A_4-ergodic component of μ. For the source's $\sigma = 4$, this is $\lambda/2$ in place of its $\lambda/3$.

Proof. For a fixed small identity neighborhood $B(b)$, the two-sided tube $a(t)B(b)a(-t) \cap a(-t)B(b)a(t)$ has off-diagonal entries $O(r)$, $r = e^{-\lambda t}$. Its diagonal entries remain near one; the determinant condition makes their product $1 + O(r^2)$. Removing a positive diagonal matrix in a fixed compact part of A_4 leaves a matrix in $B(Cr)$. A mesh in the three diagonal logarithms covers the tube by $O(r^{-3})$ translates of $B(Cr)$, with centers in a fixed compact enlargement. Domination gives tube mass $O_{\Omega,c}(r^{\sigma-3})$. ELMV Corollary 3.3 [6] takes the decay $e^{-2\eta t}$ to entropy at least η. Apply it to the constant sequence $\xi_i = \xi$, $t_i = i$, with $\eta = \lambda(\sigma - 3)/2$; enlarging the constant handles finitely many early times. For the component conclusion, if components of entropy below this bound had positive measure, a positive-measure subset would have entropy at most the bound minus some $1/k$. Its normalized mixture is dominated by a finite multiple of μ. The entropy integral formula would contradict the just-proved bound. This argument uses restricted mixtures, not domination of individual components. □

This is an exact bookkeeping consequence of an imported entropy criterion, not a new classification result. In degree four it still leaves the source's three two-block exterior-plane events to exclude.

8.1 Nonescape is a separate positive estimate

The ordinary unfolding, read with its proof in `ordinary.tex`, is

$$\int \sum_{0 \ne v \in \Lambda} f(v)\, \mathrm{d}\mu_* = \frac{1}{\sqrt{D}\,\kappa} \sum_{0 \ne \mathfrak{a} \subset R} P(\mathfrak{a}) \mathcal{V}_f(\mathrm{N}\,\mathfrak{a}/X),$$

where $\mathcal{V}_f$ sums all sixteen signs and integrates three deleted logarithmic coordinates. The local mass identity is $\sum P(\boldsymbol{l})/d(\boldsymbol{l}) = Z_q/q$; it is not the number of unit labels, which need not equal q. An upper bound for this positive vector sum in a small ball controls cusp mass by Markov and Mahler compactness, provided its arithmetic ideal-count inputs are uniform. Local Fourier energy in the exterior resolvent cannot substitute for those inputs. Likewise the source's exterior unfolding retains both traces $m, -m$, the weighted multiplicity $M(z)$ and coefficient $4/([R^\times : R^{\times,+}]\sqrt{D}\,\kappa)$. Its logarithmic Jacobian is $1/2$, offset by eight signs per trace. Under the imported planar bound, the two small product coordinates give r^4 and a summable double dyadic logarithmic weight. The small-strip remainder tends to zero with X for fixed r; one must then pass to the probability limit before sending r to zero. We make no unproved assertion for a simultaneous shrinking $r(X)$ or for a limit of mass less than one.

9 Comparison, limitations and research priorities

The primary-literature check was made on October 8, 2026, using only public titles. ELMV's cubic result [7] is a different degree-three equidistribution theorem; its validity is not a proof of

the arbitrary-order primitive quartic claim. Khayutin's double torus invariants [8] combine local arithmetic with dynamics, rather than treating a coordinate representation as a cancellation estimate. The 2026 published Wieser–Yang article [9] addresses quartic toral sets of biquadratic, cyclic or dihedral type. Those hypotheses are not the primitive A_4/S_4 case considered here. Bhargava's quartic-ring parametrization [10] supplies context for cubic resolvents but does not by itself verify this source's exterior stabilizer order or its uniform bounds. EKL classification [11] and ELMV's tube criterion remain independent dynamical inputs.

The direct public repository page was unavailable to the browsing tool during this check. The local source snapshot was read and hashed independently; the parent authenticated its equality to the captured October 8 tree. Neither repository provenance nor publisher provenance certifies a proof. The bounded search does not establish historical priority for elementary local refinements.

Advantages. The conductor exponent gives an explicit and substantially smaller support enlargement with dyadic conditions visible. Retaining the Smith mass, the Walsh coefficient norm and the discriminant-index relation avoids three unnecessary losses. The scalar formula makes the finite main-term weight exact. Orthogonal shell truncation and finite trace projection offer reproducible energy diagnostics without erasing trace-fiber normalization or arithmetic correlations.

Limitations. The period exponent is sufficient, not optimal in every lattice. The Smith bound may have no aggregate gain where only one negative direction dominates. The local L^2 constant is not a pointwise bound on arithmetic samples. The reduced cap is conditional on the source's resolvent and order-scale inputs and supplies no new global modulus range. Cubic-resolvent multiplicity, unramified weighted sieve, nonscalar congruence counting, tightness, and classification have not been replaced by the private energy framework or by finite tests.

Feasible next work. Compute conductor exponents and full Smith profiles for explicit families of quartic orders, then determine whether their distribution improves a named nonscalar branch uniformly. A second narrow task is to characterize dyadic inverse-integral vanishing on R_β; this may be more informative than a blanket absolute bound. Finally, re-run the complete exponent budget before claiming any larger additive modulus or smaller box scale. The present report stops at the proved local and expressly conditional consequences.

10 Evidence and contribution assessment

The literal-read ledger records full reads of the main file, macros, introduction, packets, local Fourier, additive section, conclusion, references and source README. Selected intervals only were read in ordinary vectors, rigidity, and exterior and sieve arguments. A separate structural search mapped every section and label. The source files were hashed before and after inspection; an unread interval is not promoted to checked merely because its label is mapped.

The private proof reads were bounded intervals of the volume-two master atoms and TR1674 originals, not only catalogue summaries. They covered Fourier normalization, cutoff energy and weighted Mellin norms, and exact finite Gram/energy identities with residual terms. No claim of whole-book or whole-private-corpus reading is made. M034's moment and entropy mechanisms are explicitly reused, with the new quartic limitations retained.

Author-run finite checks test scalar residue counts, optimized translation conditions and split nonmaximal-order examples, full Smith-index identities in finite models, trace-projection energy, and rational exponent bookkeeping. These are sanity checks, not certification of arithmetic uniformity.

Parent checks are separately attributed and excluded from author counts. Exact test counts, hashes, compiler diagnostics, final-page visual inspection and the frozen file manifest are in the accompanying owned audit records.

The assessment is C3/5: a substantive source-specific local support refinement, exact scalar normalization, orthogonal energy formulation, and conditional profile compression, with complete local proofs. It is not C4 or C5 because the central global arithmetic and dynamical obligations remain. Applicability L3 is unchanged and is a different classification from contribution strength.

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